

KIVAIRU SAMUEL

AI Engineer • Software Developer • Data Analyst • Embedded Systems Researcher

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TECHNICAL SKILLS

Programming

TypeScript • JavaScript • Python • C/C++ • SQL

Frontend

React • Next.js • Tailwind CSS • HTML5 • CSS3
• Framer Motion

Backend

Node.js • Firebase • REST APIs

AI & Data

Machine Learning • TensorFlow • PyTorch •
Pandas • NumPy • Data Visualization •
Statistical Analysis

Embedded Systems

ESP32 • STM32 • Arduino • PlatformIO • ROS2
• PX4 • Telemetry Systems

Tools

Git • GitHub • VS Code • Linux • Postman •
Figma

EDUCATION

University of Nairobi

B.Sc. in Statistics
2024 – Expected 2028

Statistical Modelling

Machine Learning

Data Science

Probability

Research Methods

PROFESSIONAL DEVELOPMENT

Power Learn Project (PLP)

Software Development Program

Python Programming

Web Development

Database Design

Software Engineering Principles

Startup & Professional Dev.

LANGUAGES

English

Kiswahili

PROFESSIONAL SUMMARY

Statistics undergraduate at the University of Nairobi with strong experience in AI engineering, full-stack software development, embedded systems, and data analytics. Passionate about building intelligent software, autonomous systems, and data-driven solutions that solve real-world problems. Experienced with React, Next.js, TypeScript, Python, Firebase, TensorFlow, and modern web technologies while actively researching autonomous flight systems and edge AI.

FEATURED PROJECTS

Disease Prediction & Forecasting AI

Machine learning platform for disease prediction and forecasting using statistical and AI models.

AgriNova AI

AI-powered agricultural assistant supporting crop management and smart farming.

Portfolio Platform

Production-ready portfolio built with Next.js 16, React 19, TypeScript, Tailwind CSS, Framer Motion, SEO optimization, analytics, and server-side rendering.

Healthcare Analytics Platform

Interactive healthcare analytics dashboards for data-driven decision making.

Autonomous Flight Controller Research

Embedded firmware research using ESP32/STM32 focused on telemetry and autonomous flight systems.

EXPERIENCE

Hardware & Systems Lead

2025 – Present

Aerospace System Research Initiative

- Researching embedded systems for autonomous aerospace applications.
- Developing telemetry and firmware prototypes.
- Working with ESP32, STM32, C/C++, and autonomous flight technologies.

Full-Stack & AI Developer

2024 – Present

Personal Engineering Projects

- Designing AI-powered web applications.
- Building modern production-ready software with React, Next.js, Firebase, and TypeScript.
- Applying statistical modelling and machine learning to real-world datasets.

ACHIEVEMENTS

- Built a production-grade engineering portfolio using Next.js 16.
- Participated in AI hackathons focused on solving real-world problems.
- Researching autonomous systems and embedded aerospace technologies.
- Combining statistics, AI, and software engineering to build intelligent systems.